

Game Analysis: The (10, 10) Rock Game

Two piles of 10 rocks each. On your turn, take 1 from Pile A, 1 from Pile B, or 1 from both. The player who takes the last rock wins. Your opponent moves first.

W-positions and L-positions

A position is a **L-position** if the player about to move loses (with optimal play), and an **W-position** if they win. The terminal position (0, 0) is a P-position since the player facing it has already lost.

The Pattern

Seeing the pattern in small cases reveals:

$$(a, b) \text{ is a L-position} \iff a \equiv 0 \pmod{2} \text{ and } b \equiv 0 \pmod{2}$$

Proof:

Both even $\Rightarrow$ every move leads to an W-position. The three moves give $(a - 1, b)$, $(a, b - 1)$, $(a - 1, b - 1)$. In each case at least one coordinate becomes odd, so no move reaches a L-position. Hence the current player loses.

At least one odd $\Rightarrow$ some move leads to a L-position.

- a odd, b even: take from A $\rightarrow (a - 1, b)$, both even.
- a even, b odd: take from B $\rightarrow (a, b - 1)$, both even.
- Both odd: take from both $\rightarrow (a - 1, b - 1)$, both even.

In every case there is a move to a L-position. Hence the current player wins.

Result for (10, 10)

Since $10 \equiv 0 \pmod{2}$, the position (10, 10) is a **L-position**. The opponent moves first and faces a L-position, so **we win** with optimal play.

Winning Strategy

After every opponent move, restore both piles to even numbers:

- Takes from A only (a odd, b even): take 1 from A $\rightarrow$ both even.
- Takes from B only (a even, b odd): take 1 from B $\rightarrow$ both even.
- Takes from both (both odd): take from both $\rightarrow$ both even.

We always hand our opponent a L-position. This continues until you take the last rock(s) at $(2, 2) \rightarrow (1, 1) \rightarrow (0, 0)$ and win.